BMW Group Standard **Released** **GS 91010-2**
2021-12

Replacement for
GS 91010-2:2021-07

Marking of parts

Marking with Data Matrix Code or RFID label

Requirements

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Continued page 2 to 37

Contents

Foreword

The German version is binding.

This Group Standard has been coordinated with the responsible departments of the BMW Group. The names of the persons creating, releasing and checking this Group Standard are filed and can be retrieved in the master data of the TEREG system.

Amendments

The following amendments have been made to GS 91010-2:2021-07:

- error corrected in the German structure: Variant 2 and variant 3 are now in separate sections;
- Group Standard editorially revised.

Previous editions

2009-10, 2016-02, 2019-09, 2020-12, 2021-07

1 Scope and purpose

In order to ensure the correct fitment and the process of traceability of single parts, part numbers and serial numbers are filed in data processing systems (e.g. IPS-Q, BVIS, SPS or similar systems). This is realized by scanning of Data Matrix Codes or RFID labels.

This Group Standard applies to all DMC and RFID tags on parts that are scanned at BMW Group.

2 Normative references

This Group Standard incorporates provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. The respective latest edition of the publication is applicable.

DID-DE-0248631[1)]	Checkliste DMC-Produktauslegung
GS 90019	Part numbers; Rules for the allocation and assignment
GS 91001	Marking of parts; With trademark and part identification data
GS 91005-3	Technical drawings; Definitions, releases, changes, design verification signatures
ISO 1073-1	Alphanumeric character sets for optical recognition; Part 1: Character set OCR-A; Shapes and dimensions of the printed image
ISO/IEC 15415	Information technology; Automatic identification and data capture techniques; test specification for bar code symbol print quality; two-dimensional symbols
ISO/IEC 16022	Information technology; International symbology specification; data matrix
ISO/IEC 18000-6	Information technology; Radio frequency identification for item management; Part 6: Parameters for air interface communications at 860 MHz to 930 MHz
ISO/IEC 18000-63	Information technology; Radio frequency identification for item management; Part 63: Parameters for air interface communications at 860 MHz to 960 MHz
ISO/IEC 29158	Information technology; Automatic identification and data capture techniques; Quality guideline for direct marking of parts (DPM)
Reference Card V5[2)]	Data Quality; V5 Reference card
VDA 5500	Basic principles for RFID application in the automotive industry

1) available in BMW Group DMS or at the respective specialist department
2) see BMW Group Partner Portal (B2B)

3 Abbreviations, terms and definitions

3.1 Abbreviations

AFI	Application Family Identifier
AI	Change index
BBG	Confirmation build
BE-Team	Modular design team
BVIS	Part variant information system
CFRP	Carbon fiber reinforced plastics
CoC CAQ	Center of Competence Computer Aided Quality
EOT	End of Text
EPC	Electronic Product Code
FZS	Vehicle Control Code

Gen5	5th Generation electric drives at BMW
Gen5+	Redesign of the 5th Generation electric drives at BMW
Gen6	6th Generation electric drives at BMW
IPM	Integrated Process Data Management
IPS-Q	International Production Steering System - Quality (IT system)
ItO	Idea to Offer
OCV	open circuit voltage
PPZ	Test plan drawing (IT system)
RFID	Radio Frequency Identification
SE-Team	Series Development Team
SPS	Programmable Logic Controller
UII	Unique Item Identifier

3.2 Terms

There are no terms in this document.

3.3 Definitions

Serial/part number	The serial or part number contains the exact date of production of a part as well as information on place of assembly and manufacturer in coded form. Together with the part number and the change index the part and its origin can be clearly identified. The specifications for serial numbers are only applicable for scopes that have not been regulated otherwise. These scopes shall be agreed separately with the IPS-Q planners of the plants.

4 Designation

There is no designation in this document.

5 Permitted symbology

The DMC according to ISO/IEC 16022 with the error correction ECC200 according to ISO/IEC 16022 shall be used.

If, in exceptional cases, another DMC is required, this shall be agreed in writing via e-mail (barcodekennung@list.bmw.com) with the "CoC CAQ".

6 Structure

6.1 Data content

In addition to the detailed definitions of contents, the general restrictions described herein apply:

- only the capital letters A-Z according to ASCII and numbers 0 to 9 are permitted as individual characters;
- special characters and spaces are explicitly prohibited;
- the lengths shall be complied with precisely, there are no exceptions allowed here. Exceptions regarding content are possible (for example: The supplier replaces the BMW serial number with his own data. However, such exceptions shall be agreed in writing via e-mail (barcodekennung@list.bmw.com) with "CoC CAQ".

Recommendations of the variant to be used for new scopes are listed in Table 1 and Table 2. Preferably, the code variants listed in Table 1 shall be used.

Table 1: Code variants and RFID variants

Vehicle plant variant	**Variant**
For assuring installation only (part number check), no part traceability required	1
For assuring installation including part traceability (serial number)	3a
Control units that have digitally recorded the 10 digit serial number	3b
Technology body shop (IPM supported part tracking)	3c
Sequence tests (direct order reference)	5
Supplier-specific data required in DMC	7
Sequence tests (direct order reference) and supplier-specific data in DMC are required	8
RFID content / DMC as optical validation of RFID label	RFID 1 RFID 1a RFID 2 RFID 2a

The code variants in Table 2 shall only be used if required.

Table 2: Further code variants

Engine assembly	M
Inhouse parts mechanical production	H
CFRP production	CFRP
Mechanical production gear set in Plant 2.1	R
High voltage battery module	HVM
High voltage battery cell up to Gen5	HVZ
High voltage battery cell as of Gen5+	HVZa
Technology body shop (PPZ supported part tracking)	K

6.1.1 Assignment of installation location code (barcode)

To ensure the clarity and uniformity of installation location identifications (barcode) for IPS-Q, they are assigned centrally by the “CoC CAQ”.

NOTE Applications for barcodes can be addressed to barcodekennung@list.bmw.com stating the name and department code of the planner responsible for the assembly process, the Unified Parts Grouping (I.P.G.), Product Process Group (PPG) and a part number example.

6.1.2 Assignment of code for manufacturer and assembly line

The identifications for manufacturer and assembly line are normally to be assigned with the value “0”. Different, specific values shall be assigned only for a part with several suppliers or assembly lines. Codes that have been assigned retain their part number specific validity.

Unless there is a risk of mix-up, it is recommended to use the first letter of the company name a supplier code.

The assignment of the identifications for supplier and assembly line at the supplier shall be carried out by the respective development department and documented on the drawing (see section 9).

6.1.3 Production counter

If an index change is required with the same serial number the day production counter shall not be restarted.

The serial number shall be unique.

6.1.4 Data Matrix Code with part number in plain text

All code variants can be combined with the part number or vehicle identification in plain text below the DMC.

NOTE The combination of the DMC with the part number or vehicle identification in clear text facilitates the visual assignment of parts by employees in the plant.

6.2 Code variants

6.2.1 Variant 1: Part number

The DMC contains the bar code and the part number with change index.

Total length: 11 characters

This variant is to be used for installation assurance only (comparison of actual part number and required part number). It is necessary to check whether variant 3a can be used to enable future part traceability.

Table 3: Data contents

Contents	**Number of digits**	**Abbreviation**	**Position in code**
Identification installation location / barcode	2	KK	1-2
Part number according to GS 90019 (alphanumeric)	7	SSSSSSS	3-9
Change index according to GS 91005-3	2	II	10-11

EXAMPLE Battery (barcode “K7”, part number “1ABCDE9”, AI “01”)

Figure 1: K71ABCDE901

6.2.2 Variant 2: Serial or part number

This variant is outdated and should no longer be used for new applications. Variant 3a is to be used instead and the part number/AI shall be filled with zeros.

6.2.3 Variant 3: Part number and serial or part number with 4 digit day counter

The DMC contains the bar code, the part number with change index and a unique serial number with 4 digit day counter.

Total length: 23 characters

Table 4: Data contents

Contents	Number of digits	Abbreviation	Position in code
Identification installation location / barcode	2	KK	1-2
Part number according to GS 90019 (alphanumeric)	7	SSSSSSS	3-9
Change index according to GS 91005-3	2	II	10-11
Production year	2	JJ	12-13
Customer	1	C	14
Day of production year	3	TTT	15-17
Assembly line/location	1	M	18
Character of manufacturer	1	H	19
Day production counter (4 digit)	4	PPPP	20-23

EXAMPLE Battery (Barcode “K7”, part number “1ABCDE9”, AI “01”, year (e.g. 2015) “15”, customer (e.g. BMW) “B”, day “001”, assembly line/location “A”, character of manufacturer (e.g. C.) “C”, production counter “0123”)

Figure 2: K71ABCDE90115B001AC0123

6.2.4 Variant 3a: Part number and serial or part number with 5 digit day counter

The DMC contains the bar code, the part number with change index and a unique serial number with 5 digit day counter.

Total length: 23 characters

Table 5: Data contents

Contents	**Number of digits**	**Abbreviation**	**Position in code**
Identification installation location / barcode	2	KK	1-2
Part number according to GS 90019 (alphanumeric)	7	SSSSSSS	3-9
Change index according to GS 91005-3	2	II	10-11
Production year	2	JJ	12-13
Character of manufacturer	1	H	14
Day of production year	3	TTT	15-17
Assembly line/location	1	M	18
Day production counter - decimal	5	PPPPP	19-23

EXAMPLE Battery (barcode “K7”, part number “1ABCDE9”, AI “01”, year (e.g. 2015) “15”, character of manufacturer “H”, day “001”, assembly line/location “M”, production counter “12345”)

Figure 3: K71ABCDE90115H001M12345

6.2.5 Variant 3b: Part number and serial or part number with 10 digit day counter

The DMC contains the bar code, the part number with change index and a unique serial number with 10 digit day counter. This variant is used for control units with 10 digit serial number in control unit.

Total length: 23 characters

Table 6: Data contents

Contents	Number of digits	Abbreviation	Position in code
Identification installation location / barcode	2	KK	1-2
Part number according to GS 90019 (alphanumeric)	7	SSSSSSS	3-9
Change index according to GS 91005-3	2	II	10-11
Character of manufacturer	1	H	12
Assembly line/location	1	M	13
Day production counter - decimal	10	PPPPPPPPPP	14-23

EXAMPLE Battery (barcode “K7”, part number “1ABCDE9”, AI “01”, character of manufacturer “H”, assembly line/location “M”, production counter “0000012345”)

Figure 4: K71ABCDE901HM0000012345

6.2.6 Variant 3c: Part number and manufacturing time

The DMC contains the bar code, the part number with change index, the manufacturer, the plant, the assembly line, the shift counter and the unique manufacturing time. The coding of the manufacturing time is described in Table A.1.

Total length: 23 characters

Table 7: Data contents

Contents	Number of digits	Abbreviation	Position in code
Identification installation location / barcode	2	KK	1-2
Part number according to GS 90019 (alphanumeric)	7	SSSSSSS	3-9
Change index according to GS 91005-3	2	II	10-11
Character of manufacturer	1	H	12
Plant	1	W	13
Assembly line/location	1	M	14
Shift counter (if required, otherwise 0)	1	S	15
Manufacturing time (see Table A.1)	8	DDDDDDDD	16-23

EXAMPLE Battery (barcode “K7”, part number “1ABCDE9”, AI “01”, character of manufacturer “H”, assembly line/location “M”, shift counter “2”, manufacturing date “GBAK5229”)

Figure 5: K71ABCDE901HMV2GBAK5229

6.2.7 Variant 4: Part number and serial or part number with 4 digit day counter and supplier number

The DMC contains the bar code, the part number with change index and a unique serial number with 4 digit day counter. This variant shall only be used if the supplier number is to be processed. Otherwise variants 3, 3a, 3b or 3c shall be used.

Total length: 31 characters

Table 8: Data contents

Contents	Number of digits	Abbreviation	Position in code
Identification installation location / barcode	2	KK	1-2
Part number according to GS 90019 (alphanumeric)	7	SSSSSSS	3-9
Change index according to GS 91005-3	2	II	10-11
Production year	2	JJ	12-13
Customer	1	C	14
Day of production year	3	TTT	15-17
Assembly line/location	1	M	18
Character of manufacturer	1	H	19
Day production counter (4 digit)	4	PPPP	20-23
Supplier number according to GS 91001	6	LLLLLL	24-29
Supplier location address according to GS 91001	2	NN	30-31

EXAMPLE Battery (Barcode “K7”, part number “1ABCDE9”, AI “01”, year (e.g. 2015) “15”, customer (e.g. BMW) “B”, day “001”, assembly line/location “A”, character of manufacturer (e.g. C.) “C”, production counter “0123”, supplier number 123456, supplier code “78”)

Figure 6: K71ABCDE90115B001AC012312345678

6.2.8 Variant 4a: Part number and serial or part number with 5 digit day counter and supplier number

The DMC contains the bar code, the part number with change index and a unique serial number with 5 digit day counter. This variant shall only be used if the supplier number is to be processed. Otherwise, variant 3a shall be used.

Total length: 31 characters

Table 9: Data contents

Contents	Number of digits	Abbreviation	Position in code
Identification installation location / barcode	2	KK	1-2
Part number according to GS 90019 (alphanumeric)	7	SSSSSSS	3-9
Change index according to GS 91005-3	2	II	10-11
Production year	2	JJ	12-13
Character of manufacturer	1	H	14
Day of production year	3	TTT	15-17
Assembly line/location	1	M	18
Day production counter - decimal	5	PPPPP	19-23
Supplier number according to GS 91001	6	LLLLLL	24-29
Supplier location address according to GS 91001	2	NN	30-31

EXAMPLE Battery (barcode “K7”, part number “1ABCDE9”, AI “01”, year (e.g. 2015) “15”, character of manufacturer “H”, day “001”, assembly line/location “M”, production counter “12345”, supplier number “123456”, supplier code “78”)

Figure7: K71ABCDE90115H001M1234512345678

6.2.9 Variant 5: Vehicle identification (sequence check)

The DMC contains a barcode and a vehicle identification. This variant is designed for parts with vehicle reference, where vehicle assignment is already clear when the barcode is applied (sequence harmonization).

Total length: 9 characters

Table 10: Data contents

Contents	Number of digits	Abbreviation	Position in code
Identification installation location / barcode	2	KK	1-2
The vehicle identification may include the order number, the 7 digit VIN or the vehicle steering code (FZS)	7	IIIIIII	3-9

EXAMPLE Tank (barcode "TA", vehicle order number "1234567")

Figure 8: TA1234567

6.2.10 Variant 6: Part number, vehicle identification

The DMC contains the bar code, the part number, the change index and the vehicle identification. This variant is designed for parts with vehicle reference, where vehicle assignment is already clear when the barcode is applied, but where part number information is required as well.

Total length: 18 characters

Table 11: Data contents

Contents	Number of digits	Abbreviation	Position in code
Identification installation location / barcode	2	KK	1-2
Part number according to GS 90019 (alphanumeric)	7	SSSSSSS	3-9
Change index according to GS 91005-3	2	II	10-11
The vehicle identification may include the order number, the 7 digit VIN or the vehicle steering code (FZS)	7	IIIIIII	12-18

EXAMPLE Fuel tank (Barcode “TA”, part number “1ABCDE9”, AI “01”, vehicle VIN “AB08152”)

Figure 9: TA1ABCDE901AB08152

6.2.11 Variant 7: Part number, serial or part number with 5 digit day counter, supplier number and supplier-specific data

The DMC contains the bar code, the part number with change index and a unique serial number with 5 digit day counter and supplier-specific data. This variant is to be used if the supplier wants to include his own data in the DMC. If the supplier or serial number is not required, these data shall be filled with "0". The total length of 60 characters is to be observed unconditionally! The restrictions regarding the number of permitted characters as described in section 6.1 are also applicable in this case.

Total length: 60 characters

Table 12: Data contents

Contents	**Number of digits**	**Abbreviation**	**Position in code**
Identification installation location / barcode	2	KK	1-2
Part number according to GS 90019 (alphanumeric)	7	SSSSSSS	3-9
Change index according to GS 91005-3	2	II	10-11
Production year	2	JJ	12-13
Character of manufacturer	1	H	14
Day of production year	3	TTT	15-17
Assembly line/location	1	M	18
Day production counter - decimal	5	PPPPP	19-23
Supplier number according to GS 91001	6	LLLLLL	24-29
Supplier location address according to GS 91001	2	NN	30-31
Supplier-specific data: The data shall correspond to the specifications in section 6.1 and always be filled to the required length. It is irrelevant whether it is left or right-justified and which characters are used to fill. "0" is recommended (numeric zero).	29	XXXXXXXXXX XXXXXXXXXX XXXXXXXXX	32-60

EXAMPLE Battery (Barcode "K7", part number "1ABCDE9", AI "01", year (e.g. 2015) "15", customer (e.g. BMW) "B", day "001", assembly line/location "A", character of manufacturer (e.g. C.) "C", production counter "12345", supplier number "123456", supplier code "78" , supplier-specific data "X10156713ZA000000000000000000")

Figure 10: K71ABCDE90115C001A1234512345678X10156713ZA000000000000000000

6.2.12 Variant 8: Vehicle identification (sequence check) and supplier-specific part

The DMC contains a barcode, a vehicle identification and supplier-specific data. This variant is designed for parts with vehicle reference, where vehicle assignment is already clear when the barcode is applied (sequence harmonization). In addition, it is possible to include supplier-specific data in the DMC.

Total length: 31 characters

Table 13: Data contents

Contents	Number of digits	Abbreviation	Position in code
Identification installation location / barcode	2	KK	1-2
The vehicle identification may include the order number, the 7 digit VIN or the vehicle steering code (FZS)	7	IIIIIII	3-9
Supplier-specific data: The data shall correspond to the specifications in section 6.1 and always be filled to the required length. It is irrelevant whether it is left or right-justified and which characters are used to fill. “0” is recommended (numeric zero).	22	XXXXXXXXXX XXXXXXXXXX XX	10-31

EXAMPLE Center console (bar code “MK”, vehicle order number “1234567”, supplier-specific data “X10156713ZA00000000000”)

Figure 11: MK1234567X10156713ZA00000000000

6.3 RFID variants

6.3.1 General requirements

When using RFID labels, one of the following RFID variants shall be selected depending on whether it is a part with part number and serial number (variant RFID 1 or RFID 1a) or with vehicle identification and serial number (variant RFID 2 or RFID 2a). The data content in the RFID label and the data content of the DMC of the variants RFID 1 and RFID 2 are identical except for the data identifier. With the variants RFID 1a and RFID 2a the DMC contains a supplier specific part in addition. The RFID variants shall also be used if the DMC is not printed directly on the RFID label (e.g. when a DMC label and an RFID label are used).

The day production counter shall be filled hexadecimally (0 to F). This enables a maximum output of 4 095 parts per day.

Type of RFID label memory allocation:

The data is stored on the logical memory bank MB01 (EPC) of the RFID label. This memory bank contains the header (control information) and the unique item identifier (UII) (corresponds to the data content). This memory bank shall be assigned according to the variant described in VDA 5500 (ISO). If the last word in the EPC is not completely filled with the required data content, padding bits are appended to fill up to full 16-bit words (analogous to VDA 5500). The command EOT is not used

The following requirements concerning the data content of the memory bank MB01 shall be observed:

- Header:
  - Toggle-Bit = 1;
  - AFI = A1 (parts without hazardous goods identifier) or A4 (parts with hazardous goods identifier);
- UII: Coding of contents according to VDA 5500 ("Coding table 6-bit Encoding");
- Data identifier: The value "S" shall always be used as data identifier for parts.

In general, the information in the UII shall be provided with permanent write protection after successful coding. Exceptions shall be agreed with BMW within the scope of SE teamwork.

6.3.2 Variant RFID 1: Part number, serial number

The variant RFID 1 contains the data identifier (only in RFID), the barcode, the part number and the unique serial number.

Total length: 21 characters RFID / 20 characters DMC

Table 14: Data contents

Contents	Number of digits	Abbreviation	Position in RFID code	Position in DMC code
Data Identifier	1	D	1	non-existent
Identification installation location / barcode	2	KK	2-3	1-2
Part number according to GS 90019 (alphanumeric)	7	SSSSSSS	4-10	3-9
Change index according to GS 91005-3	2	II	11-12	10-11
Manufacturing time[1)]	3	DDD	13-15	12-14
Character of manufacturer	1	H	16	15
Assembly line/location	1	M	17	16
Day production counter (hexadecimal)	3	PPP	18-20	17-19
Quality characteristic[2)]	1	Q	21	20

1) The coding of the manufacturing time is described in Table A.1. Only the first three digits (year, month, day) shall be used.
2) The quality characteristics are described in Table A.2.

EXAMPLE Battery (Barcode “K7”, part number “1ABCDE9”, AI “01”, manufacturing time (e.g. 10.11.2019) “JBA”, character of manufacturer “H”, assembly line/location “0”, day production counter (e.g. 1310) “51E”, quality

Figure 12: K71ABCDE901JBAH051E1

characteristic (part with quality control loop) “1”)

6.3.3 Variant RFID 1a: Part number, serial number and supplier-specific part in DMC

The variant RFID 1a contains the data identifier (only in RFID), the barcode, the part number with change index (AI), the unique serial number and a supplier-specific part (only in DMC).

Total length: 21 characters RFID / 59 characters DMC

Table 15: Data contents

Contents	**Number of digits**	**Abbreviation**	**Position in RFID code**	**Position in DMC code**
Data Identifier	1	D	1	non-existent
Identification installation location / barcode	2	KK	2-3	1-2
Part number according to GS 90019 (alphanumeric)	7	SSSSSSS	4-10	3-9
Change index according to GS 91005-3	2	II	11-12	10-11
Manufacturing time[1)]	3	DDD	13-15	12-14
Character of manufacturer	1	H	16	15
Assembly line/location	1	M	17	16
Day production counter (hexadecimal)	3	PPP	18-20	17-19
Quality characteristic[2)]	1	Q	21	20
Supplier-specific data: The data shall correspond to the specifications in section 6.1 and always be filled to the required length. It is irrelevant whether it is left or right-justified and which characters are used to fill. “0” is recommended (numeric zero).	39	XXXXXXXXXXX XXXXXXXXXXX XXXXXXXXXXX XXXXXXX	non-existent	21-59

1) The coding of the manufacturing time is described in Table A.1. Only the first three digits (year, month, day) shall be used.
2) The quality characteristics are described in Table A.2.

EXAMPLE Battery (Barcode “K7”, part number “1ABCDE9”, AI “01”, manufacturing time (e.g. 10.11.2019) “JBA”, character of manufacturer “H”, assembly line/location “0”, day production counter (e.g. 1310) “51E”, quality characteristic (part with quality control loop) “1”, supplier-specific data “10156713ZA0000000000000000000000000000”)

Figure 13: K71ABCDE901JBAH051E1X10156713ZA0000000000000000000000000

6.3.4 Variant RFID 2: Vehicle identification (sequence check) with serial number

The variant RFID 2 contains the data identifier (only in RFID), the barcode, the vehicle identification and the unique serial number.

Total length: 19 characters RFID / 18 characters DMC

Table 16: Data contents

Contents	**Number of digits**	**Abbreviation**	**Position in RFID code**	**Position in DMC code**
Data Identifier	1	D	1	non-existent
Identification installation location / barcode	2	KK	2-3	1-2
Vehicle identification (may contain order number, 7 digit vehicle identification number or vehicle steering code (FZS)	7	IIIIIII	4-10	3-9
Manufacturing time[1)]	3	DDD	11-13	10-12
Character of manufacturer	1	H	14	13
Assembly line/location	1	M	15	14
Day production counter (hexadecimal)	3	PPP	16-18	15-17
Quality characteristic[2)]	1	Q	19	18

1) The coding of the manufacturing time is described in Table A.1. Only the first 3 digits (year, month, day) shall be used.
2) The quality characteristics are described in Table A.2.

EXAMPLE Fuel tank (Barcode “TA”, vehicle order number “1234567”, manufacturing time (e.g. 10.11.2019) “JBA”, character of manufacturer “H”, assembly line/location “0”, production counter (e.g. 1310) “51E”, quality characteristic (part with quality control loop) “1”)

Figure 14: TA1234567JBAH051E1

6.3.5 Variant RFID 2a: Vehicle identification (sequence check) with serial number and supplier-specific part in DMC

The variant RFID 2a contains the data identifier (only in RFID), the barcode, a vehicle identification and the unique serial number and a supplier-specific part (only in DMC).

Total length: 19 characters RFID / 57 characters DMC

Table 17: Data contents

Contents	Number of digits	Abbreviation	Position in RFID code	Position in DMC code
Data Identifier	1	D	1	non-existent
Identification installation location / barcode	2	KK	2-3	1-2
Vehicle identification (may contain order number, 7 digit vehicle identification number or vehicle steering code (FZS)	7	IIIIIII	4-10	3-9
Manufacturing time 1)	3	DDD	11-13	10-12
Character of manufacturer	1	H	14	13
Assembly line/location	1	M	15	14
Day production counter (hexadecimal)	3	PPP	16-18	15-17
Quality characteristic 2)	1	Q	19	18
Supplier-specific data: The data shall correspond to the specifications in section 6.1 and always be filled to the required length. It is irrelevant whether it is left or right-justified and which characters are used to fill. "0" is recommended (numeric zero).	39	XXXXXXXXXXX XXXXXXXXXXX XXXXXXXXXXX XXXXXXX	non-existent	19-57

1) The coding of the manufacturing time is described in Table A.1. Only the first 3 digits (year, month, day) shall be used.
2) The quality characteristics are described in Table A.2.

EXAMPLE Fuel tank (Barcode "TA", vehicle order number "1234567", manufacturing time (e.g. 10.11.2019) "JBA", character of manufacturer "H", assembly line/location "0", production counter (e.g. 1310) "51E", quality characteristic (part with quality control loop) "1", supplier-specific data "X10156713ZA0000000000000000000000000000")

Figure 15: TA1234567JBAH051E1X10156713ZA0000000000000000000000000000

6.4 Further code variants

6.4.1 Variant M

DMC variant M is used exclusively in engine assembly plants. This variant is not readable in the vehicle plants.

The number of code elements in variant M shall amount to 18 x 18 elements.

A different number of code elements shall be agreed with the respective process planning department.

Table 18: Data content for DMC variant M for single part tracking

Contents	**Number of digits**	**Format**	**Position in code**[2)]
Part number according to GS 90019 (alphanumeric)	7	SSSSSSS	1-7
Change index according to GS 91005-3	2	II	8-9
Supplier number of supply plant	8	12345678	10-17
Date	6	JJMMTT	18-23
Day production counter[1]	5	12345	24-28

1) Starts with 1 in ascending order at the respective change of the date.
2) The unused digits can be used for other purposes.

EXAMPLE (part number/AI “1ABCDE901”, supplier number “12345678”, date “151201”, counter “00123”)

Figure 16: 1ABCDE90112345678151201 00123

Table 19: Data content for DMC variant M for line-related batch tracing

Contents	**Number of digits**	**Format**	**Position in code**[2)]
Part number / selection table number according to GS 90019 (alphanumeric)	7	SSSSSSS	1-7
Change index according to GS 91005-3	2	II	8-9
Supplier number of supply plant	8	12345678	10-17
Date	6	JJMMTT	18-23
Number of KLT or deep drawing foil[1)]	5	12345	24-28
Identification of control technology (always the letter C)[3)]	1	C	29

1) Every insert / container within a load unit shall be provided with a unique number.
2) The unused digits can be used for other purposes.
3) The letter C is always required on digit 29 for identification in the processing software.

6.4.2 Variant H

DMC variant H are used exclusively for inhouse parts in mechanical production. This variant is not readable in the vehicle plants.

The number of code elements in variant H shall amount to 16 x 16 elements. With alphanumeric part numbers, 18 x 18 elements are also possible.

A different number of code elements shall be agreed with the respective process planning department.

Table 20: Data content for DMC variant H

Contents	Number of digits	Format	Position in code
Assembly line code [1)]	2	12	1-2
Date	6	JJMMTT	3-8
Day production counter[2)]	5	12345	9-13
Part number with change index	9	1ABCDE901	14-22
Assembly code[3)]	2	12	23-24

1) The production line code ensures the traceability of the part corresponding to the production line. Each production line has its own code.
2) Starts with 1 in ascending order at the respective change of the date.
3) The assembly code serves for identification of the part for assembly.

EXAMPLE (assembly line code “12”, date “151201”, counter “00123”, part number with AI “1ABCDE912”, assembly code “42”)

Figure 17: 12151201001231ABCDE91242

6.4.3 Variant CFRP

DMC of the variant CFRP are used exclusively in CFRP production. This variant is not readable in the vehicle plants.

The number of code elements in variant CFRP shall amount to 16 x 16 elements.

A different number of code elements shall be agreed with the respective process planning department.

Table 21: Data content for DMC variant CFRP

Contents	Number of digits	Format	Position in code
Part number with change index	9	SSSSSSSSS	1-9
unique consecutive number	8	12345678	10-17

EXAMPLE (Part number with AI “881008503”, unique consecutive number “22222280”)

Figure 18: 88100850322222280

6.4.4 Variant R

DMC of variant R are used exclusively for mechanical production of gear sets in Plant 2.1.

The number of code elements in variant R shall amount to 14 x 14 elements.

Total length: 10 characters

Table 22: Data content for DMC variant R

Contents	Number of digits	Abbreviation	Position in code
Part identification	1	K	1
Facility identification	2	MM	2-3
consecutive number	7	PPPPPPP	4-10

EXAMPLE Drive pinion (part identification “A”, facility identification “13”, consecutive number “1447492”)

Figure 19: A131447492

6.4.5 Variant HVM: High voltage battery module

DMC variant HVM are used exclusively in module assembly. If additional information is required in the DMC, use variant 7 for modules. The variant HVM is not suitable for IPS-Q and is not readable in vehicle plants.

The number of code elements in variant HVM shall amount to 18 x 18 elements.

Total length: 28 characters

Table 23: Data content for DMC variant HVM for single part tracking

Contents	Number of digits	Format	Position in code[2)]
Part number according to GS 90019 (alphanumeric)	7	SSSSSSS	1-7
Change index according to GS 91005-3	2	II	8-9
Supplier number of supply plant/supplier	8	12345678	10-17
Date	6	JJMMTT	18-23
Assembly line/string/location	1	M	24
Day production counter[1]	4	1234	25-28

1) Starts with 1 in ascending order at the respective change of the date.
2) The unused digits can be used for other purposes.

EXAMPLE (part number “1ABCDE9”, AI “01”, supplier number “12345678”, date “151201”, assembly line/ line “1” counter “00123”)

Figure 18: 1ABCDE9011234567815120110123

6.4.6 Variant HVZ: High voltage battery cell up to Gen5

DMC of variant HVZ are used exclusively for battery cells in high voltage batteries and module assembly up to Gen5. The variant HVZ is not suitable for IPS-Q and is not readable in vehicle plants.

The number of code elements in variant HVZ shall amount to 18 x 18 elements.

Total length: 36 characters DMC

Table 24: Data content for DMC variant HVZ for single part tracking

Contents	Number of digits	Format	Position in code[2)]
Part number according to GS 90019 (alphanumeric)	7	SSSSSSS	1-7
Change index according to GS 91005-3	2	II	8-9
Supplier number of supply plant/supplier	8	12345678	10-17
Date from start of first electrolyte filling	6	JJMMTT	18-23
Day production counter[1]	5	12345	24-28
Day after date on which the no-load voltage was measured in the end-of-line test.	3	TTT	29-31
measured 1 000-fold no-load voltage (OCV) of the cell	4	VVVV	32-35
Assembly line/location	1	M	36

1) Starts with 1 in ascending order at the respective change of the date.
2) The unused digits can be used for other purposes.

EXAMPLE (Part number “1123459”, AI “01”, supplier number “12345678”, date “151201”, counter “00123”, day after cell assembly “321”, no-load voltage (e.g. 4,123 V) “4123”, assembly line/line “1”)

Figure 20: 112345901123456781512010012332141231

6.4.7 Variant HVZa: High voltage battery cell as of Gen5+

DMC of variant HVZa are used exclusively for battery cells in high voltage batteries and module assembly for Gen5+, Gen6 and following. Variant HVZa is suitable for IPS-Q and is readable in vehicle plants.

The number of code elements in variant HVZa shall amount to 20 x 20 elements.

Total length: 30 characters DMC

Table 25: Data contents

Contents	Number of digits	Abbreviation	Position in code
Identification installation location / barcode	2	KK	1-2
Part number according to GS 90019 (alphanumeric)	7	SSSSSSS	3-9
Change index according to GS 91005-3	2	II	10-11
Production year	2	JJ	12-13
Character of manufacturer	1	H	14
Date from start of first electrolyte filling	3	TTT	15-17
Assembly line/location	1	M	18
Day production counter – alphanumeric[1], if one line can produce more than 99999 cells per day, otherwise numeric	5	PPPPP	19-23
Day after date on which the no-load voltage was measured in the end-of-line test.	3	TTT	24-26
measured 1 000-fold no-load voltage (OCV) of the cell	4	VVVV	27-30
1) Starts with 1 in ascending order at the respective change of the date. After 9 comes A to Z without umlauts per digit.			

EXAMPLE Battery (barcode “K7”, part number “1ABCDE9”, AI “01”, year (e.g. 2015) “15”, character of manufacturer “H”, day “001”, assembly line/location “M”, counter “ABCDE”, day after cell assembly “123”, no-load voltage (e.g. 4, 123V) “4123”)

Figure 20: T51ABCDE90115H001MABCDE1234123

6.4.8 Variant K: Part identification, check digits

The DMC (ECC200) contains one part identification and 2 check digits. The part identification and the check digits are separated by a blank, contrary to section 6.1.

Total length: 11 characters

Table 26: Data contents

Contents	Number of digits	Abbreviation	Position in code
Part ID[1)]	8	BBBBBBBB	1-8
Blank	1	L	9
Check digits[2)]	2	ZZ	10-11

1) The 8 digit alphanumeric part ID has different characteristics:
   – Code type IG front end IGEF number: In this case the part ID corresponds to the standardized IGEF number;
   – Code type PN floor panel part number: In this case the part ID corresponds to a unique serial number;
   – Code type K1 trunk lid: In this case the part ID corresponds to a unique manufacturing time (accurate to the second);

2) The check digits are computed automatically at the embossing points and added to the part ID. When manual inputs are made at the operating points, the check digit prevents data input errors.

EXAMPLE Code type IG front end (part ID (e.g. IGEF number) “AB123B12” , blank “ ”, check digits “02”)

Figure 21: AB123B12 02

7 Quality requirements

7.1 General requirements

The use of DMC and RFID shall be coordinated in the responsible SE/BE team at BMW.

The characteristics of the DMC shall be uniform per part type (e.g. all eccentric shafts) for all part generations and across all suppliers.

Different solutions shall be agreed by the SE/BE team and the responsible assembly planning department and documented.

NOTE The "DMC Product Design" checklist (DID-DE-0248631) serves as a tool for determining and recording the DMC specifications.

The readability of the DMC is to be verified as part of the initial sample inspection by the supplier via test report to be at least class B or 3 (good) for printed DMC or at least with class C or 2 (satisfactory) for DMC applied directly to the part in accordance with ISO/IEC 29158 or ISO/IEC 15415. The regulations in ISO/IEC 29158 are only applicable if the DMC is applied directly onto the part. For RFID labels, the correct data content on the chip shall be verified during initial sample inspection. The test report shall be filed until the valid retention period has expired.

In addition, the supplier shall provide retention samples. It is recommended to store retention samples also at BMW.

The location where the DMC/RFID labels are applied shall be agreed between the responsible development department and process planning within the plant and in the case of painted mounting parts (e.g. doors, lids) additionally with the Technology Surface. It shall be selected so as to ensure easy access by the reading device and the legibility of the DMC/RFID label in particular if applied on curved surfaces or complex part geometries even in installed condition. Deviations shall be agreed in the SE/BE team. The position of the DMC/RFID label shall be validated and optimized regarding readability in installed condition via the ItO build phases as of BBG.

If the DMC is applied on a curved surface the radian measure of the DMC shall not exceed 5 % of the circumference and 16 % of the diameter, respectively.

Final acceptance shall be carried out on site in installation condition together with process planning. A DMC is only considered to be "readable" if the BMW internal hardware captures the correct data without restrictions under local conditions in the series production process.

Table 27: General requirements

Verifiability for printed DMC	**At least class B / 3 (good) according to ISO/IEC 15415**
Verifiability for directly applied DMC	**At least class C / 2 (satisfactory) according to ISO/IEC 29158**
Minimum module thickness	0,5 mm[1)], in case of camera tests 0,8 mm
Coding mode	ASCII
Quiet zone	as per ISO/IEC 16022
Clear text lettering[2)]	Font Helvetica regular or according to ISO 1073-1
Permitted deviation in position	max. 10 % of code size in all directions[3)]
Printing contrast	min. 55 %[4)]
Geometry	square; rectangular subject to agreement with SE/BE team

1) In coordination with the “CoC CAQ” at BMW, a smaller module thickness can be used, contingent upon process-reliable readability.
2) At least part number and change index shall be indicated as clear text.
3) Other position deviations shall be coordinated with the development department responsible and process planning at BMW.
4) deviating from ISO/IEC 29158

7.2 Requirements for printed DMC and RFID labels

Adhesive labels shall be used, which guarantee readability of the data until delivery of the vehicle.

7.3 Requirements for DMC directly applied onto the part

If the material of the part is suitable, the DMC may be applied directly onto the part.

As a rule, readability of lasered DMC can only be ensured by a high-quality camera system. Therefore, the use of DMC applied directly onto the part shall be coordinated with responsible process planning department at BMW and, additionally, with the Technology Surface at BMW. The DMC shall be free from contamination, for example:

- detergent residues;
- powder traces (in case of laser marking);
- ink residues (in case of printed DMC).

If corrosion protection agents or oils are used, the readability of the DMC in the series delivery condition shall be verified between BMW production planning and the part supplier.

For needled DMC, the dot size shall be 70 % to 95 % of the cell size (image quality).

8 Quality requirements for RFID labels

8.1 General requirements

The air interface of passive RFID transponders shall comply with the requirements of ISO/IEC 18000-63/EPC Class1 Gen2. Broadband RFID labels (860 to 960) MHz shall be used. Country-specific documents are required as proof of conformity (e.g. CE).

A suitable RFID label shall be selected depending on the characteristics of the part (e.g. geometry, material, function) and installation location in the vehicle. The position of the RFID label on the part shall comply with the requirements in section 8.3 (range of readability and reading quality).

8.2 Specified standard RFID labels

Depending on the application case, the following RFID labels with corresponding BMW part number, which comply with all BMW requirements shall be used.

RFID LABEL NON-METAL: Part number: 5A15353

RFID LABEL NON-METAL STRONG: Part number: 5A3F2D0

RFID LABEL FLAG: Part number 5A15354

RFID LABEL LOOP: Part number 5A15355

RFID LABEL LOOP L2: Part number 5A3F383

RFID LABEL LOOP L3: Part number 5A3F399

If the use of the specified RFID label is not possible (e.g. due to inadequate reading range or package space reasons), an alternative RFID label shall be defined with the SE team. The selected RFID label shall comply with the requirements in section 8.3.

8.3 Technical requirements

Frequency	**(860 to 960) MHz (UHF)**
Energy	Smart Label without battery. Energy provided via the antenna (passive label)
Chip (type / manufacturer)	NXP UCODE 8 or similar (read sensitivity -21 dBm or better)
Type	EPC Class 1 Gen2
Memory	minimum 128 Bit EPC
encryption	not required
write protection	yes
Protocol	ISO/IEC 18000-6 Type C
Application	Self-adhesive label with PET substrate Stick on in one step without blisters
Print	The printing ink shall not show any quality loss (under all conditions)
Temperature range of labels	-40 °C to 150 °C
Operating temperature RFID-Inlay	-40 °C to 85 °C
Quality	A 100 % OK rate in terms of data content and readability shall be guaranteed for all applied RFID labels. For this purpose the functional capability of RFID labels shall be checked in standard RFID printers during the printing process and optionally an end-of-line check may be carried out.
Reading range	The label, applied to the part, free-standing in a room shall be readable constantly from ≥ 3 m all around.

9 Documentation

9.1 Drawing entry

The position of the DMC shall be indicated and dimensioned including tolerances in the drawing (as a rule, related to the zero point of the coordinate system of the part). The drawing entry is realized in the respective view with a reference line to a marked surface. In addition to the standard number, also the abbreviation "DMC" as marking for the data matrix code, the selected code variant and the barcode, as well as the appropriate identification in the event of multiple manufacturers and/or assembly lines shall be indicated (see section 6.1.2). DMC size and type of application (adhesive label, lasered, printed, dot peened) shall be indicated.

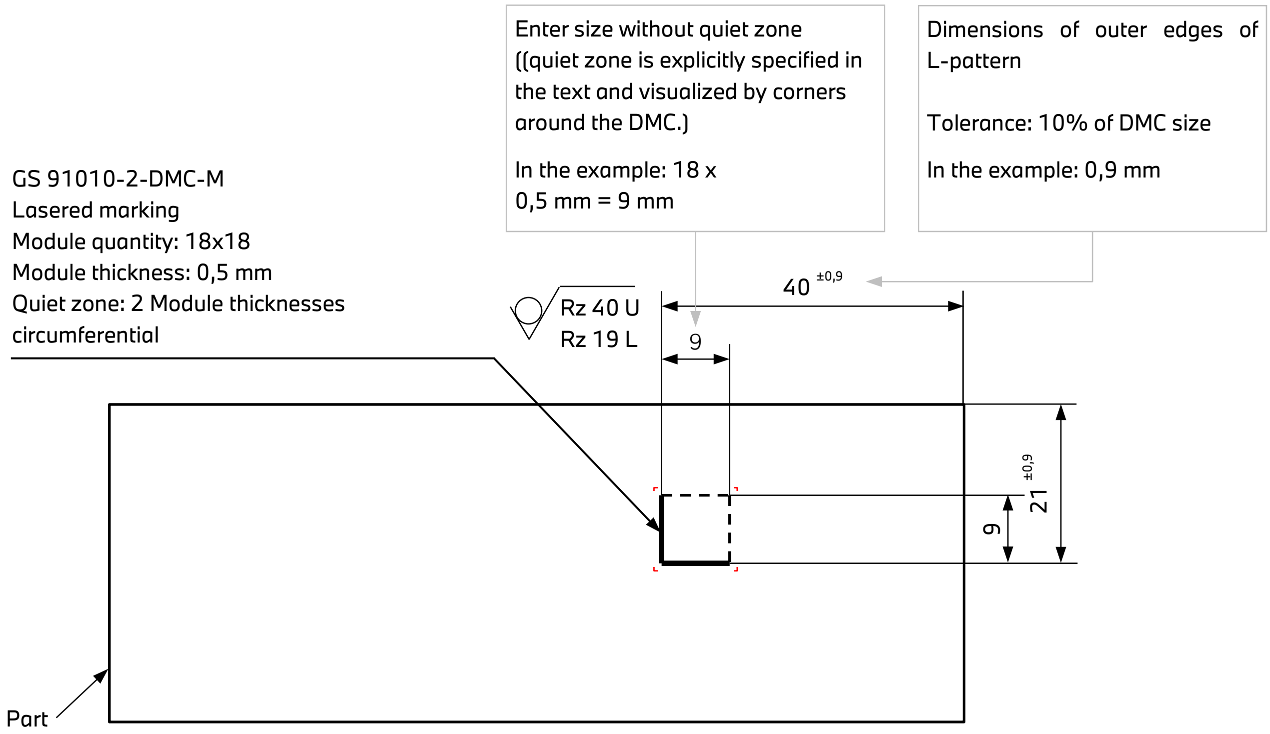

Figure 22: Example DMC variant M

Drawing entries for DMC shall be based on GS 91010-2. At least the module thickness and the quiet zone and optionally the module count and surface roughness incl. quiet zone shall be indicated. Additionally, the position of the finder pattern (L-pattern) shall be marked (see Figure 22).

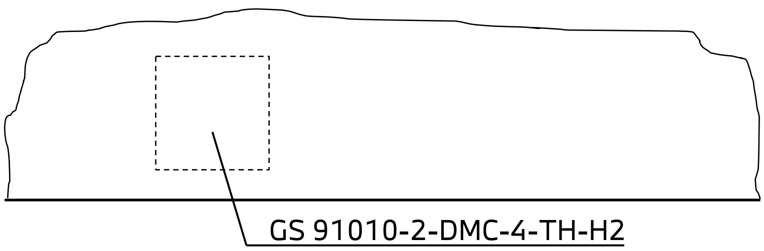

Figure 23: Example DMC variant 4 (simplified representation)
barcode TH (transmission wiring harness)
Manufacturer identification H (Manufacturer: Fa. H.)
assembly line 2 (Plant Stuttgart, line 2)

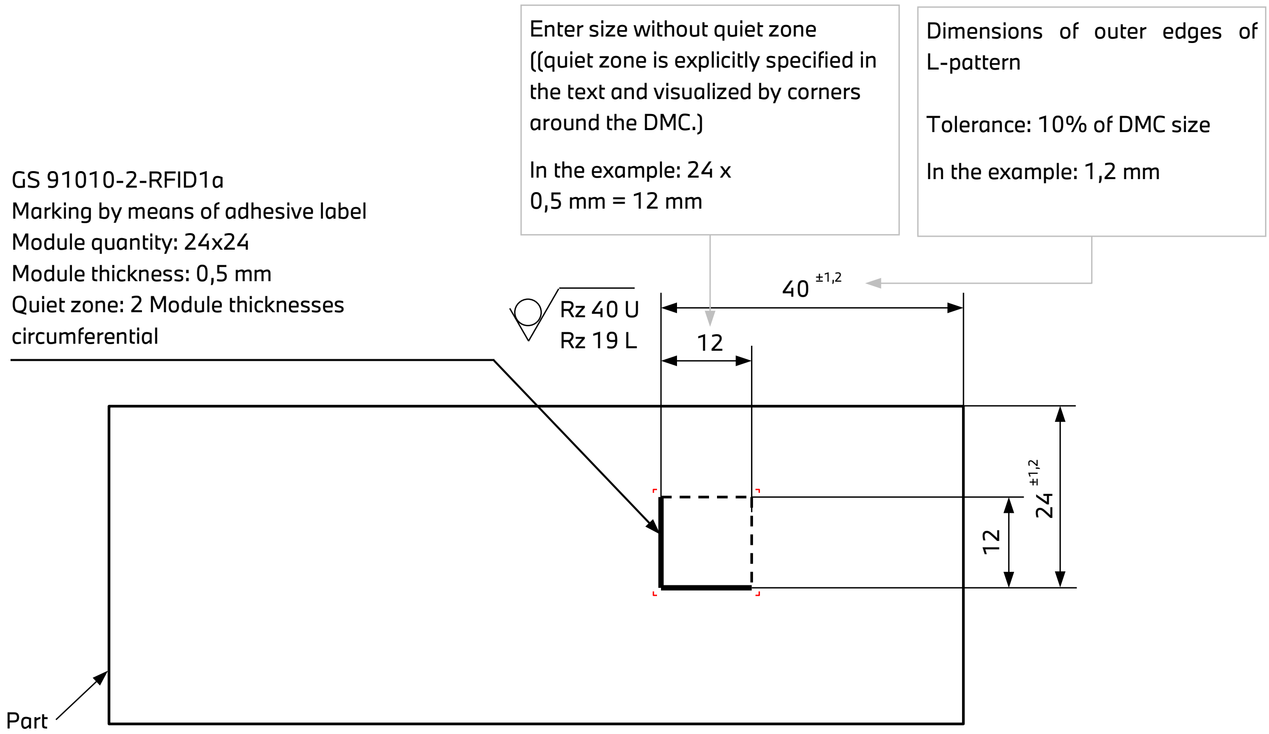

Figure 24: Example DMC Variant RFID1a

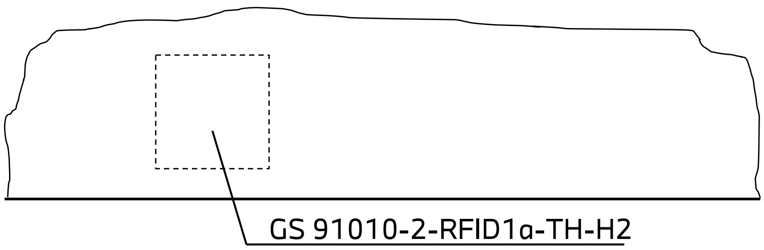

Figure 25: Example DMC RFID1a variant (simplified representation)
barcode TH (transmission wiring harness)
Manufacturer identification H (Manufacturer: Fa. H.)
assembly line 2 (Plant Stuttgart, line 2)

9.2 Drawingless data entry (ZFE)

The position of the DMC shall be indicated in the 3D model.

NOTE Additional details, e.g. exact size, position, module count and module thickness may be described.

With the introduction of the drawing-free process with the “CoC body and equipment”, the identification of the marking of parts shall be stored in the 3D release model “FRGMOD”. This can be done manually or with the “MRI tool” (“machine-readable information”, with this tool, machine-readable texts are stored that can be read out in subsequent processes using software) available within CATIA. The text shall be stored under “View additional information” (see Figure 26). The text structure is shown in Figure 27.

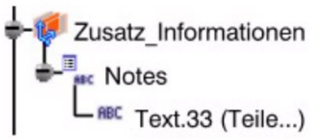

Figure 27: Text “marking of parts” in “AnnotationView” auxiliary information

Part identification marking;GS91010-DMC-4-TH-H2;Remark

Reserved word | Content of the Group Standard | Optional text for remarks

Figure 26: Description of text string “marking of parts”

The text is associatively linked with the rectangle for positioning (see Figure 28). The design of the rectangle for positioning the marking of parts is described in the Reference Card V5.

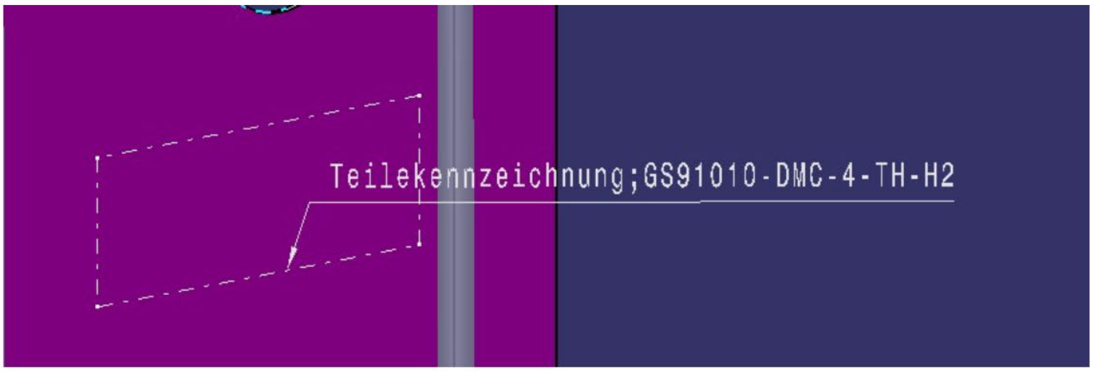

Figure 28: Text “marking of parts” in 3D model

A (Annex normative) Coding of production time

Table A.1: Coding of manufacturing time

Year	Code	Month	Code	Day	Code	Hour	Code	Minute		Second	
								1. digit	2. digit	1. digit	2. digit
2000	0	0	0	0	0	0	0	0	0	0	0
2001	1	Jan	1	1	1	1	1	1	1	1	1
2002	2	Feb	2	2	2	2	2	2	2	2	2
2003	3	Mar	3	3	3	3	3	3	3	3	3
2004	4	Apr.	4	4	4	4	4	4	4	4	4
2005	5	May	5	5	5	5	5	5	5	5	5
2006	6	June	6	6	6	6	6		6		6
2007	7	July	7	7	7	7	7		7		7
2008	8	Aug	8	8	8	8	8		8		8
2009	9	Sep.	9	9	9	9	9		9		9
2010	A	Oct	A	10	A	10	A				
2011	B	Nov	B	11	B	11	B				
2012	C	Dec	C	12	C	12	C				
2013	D			13	D	13	D				
2014	E			14	E	14	E				
2015	F			15	F	15	F				
2016	G			16	G	16	G				
2017	H			17	H	17	H				
2018	I			18	I	18	I				
2019	J			19	J	19	J				
2020	K			20	K	20	K				
2021	L			21	L	21	L				
2022	M			22	M	22	M				
2023	N			23	N	23	N				
2024	O			24	O	24	O				
2025	P			25	P						
2026	Q			26	Q						
2027	R			27	R						
2028	S			28	S						
2029	T			29	T						
2030	U			30	U						
2031	V			31	V						
2032	W										
2033	X										
2034	Z										

EXAMPLE Production time October 12, 2019 10:25:34 is coded “JACA2534”

B (Annex normative) Quality characteristics

The quality characteristic serves to illustrate the quality control loops at the supplier on the part or the test results of the part.

The characteristic has the default value “0”.

If quality control loops on the part have been agreed between BMW and the supplier (Focus ItO build phases), the code values (1 to F) can be used.

If tests on the part (e.g. tests on high voltage parts) have been agreed between BMW and the supplier, the code values (G or H) can be used.

Table A.2: Coding of quality characteristic

Quality characteristic	**Code**
No attribute (default value)	0
1. Quality control loop	1
2. Quality control loop	2
3. Quality control loop	3
4. Quality control loop	4
5. Quality control loop	5
6. Quality control loop	6
7. Quality control loop	7
8. Quality control loop	8
9th Quality control loop	9
10th Quality control loop	A
11th Quality control loop	B
12th Quality control loop	C
13th Quality control loop	D
14th Quality control loop	E
15th Quality control loop	F
Test OK	G
Test not OK	H